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Growth of Grafted Cucumber Seedlings as Affected by Different EC Levels and Cultivars

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Abstract. The nutrient solution used for plant growth contains water and essential inorganic nutrients. The concentration of these nutrients can be measured through electrical conductivity. It is essential to find the proper nutrient solution concentration because different crops exhibit different growth responses depending on the level of the electrical conductivity (EC). Therefore, this experiment was conducted to determine the proper EC level for grafted cucumber seedling production, considering different cultivars. Three cultivars of cucumber (*Cucumis sativus* L.) were used as scions for grafting: 'NakWonSeongcheongjang (NWS)', 'Sinsedae (SD)', and 'Good-morning baekdadagi (GB)'. Figleaf gourd (*Cucurbita ficifolia*) 'Heukjong' was used as the rootstock. They were grafted on September 18, 2023. After graft-taking, cucumber seedlings in plug trays were moved to a closed-type plant production system. Cucumber seedlings in plug trays were treated with the EC 1.5, 2.0, and 2.5 $dS \cdot m^{-1}$. EC of tap water was used as the control. Fresh and dry weights of shoots were significantly increased in all cultivars at the EC 2.0 and 2.5 $dS \cdot m^{-1}$ in all cultivars. In particular, leaf length, leaf width, and leaf area of NWS and GB were significantly increased in EC 2.0 and 2.5 $dS \cdot m^{-1}$. Seedling qualities such as compactness and leaf area rate were also increased in EC 2.0 and 2.5 $dS \cdot m^{-1}$. As the EC levels increased, specific leaf area increased. SPAD, chlorophyll a and b were significantly higher in EC 2.0 and 2.5 $dS \cdot m^{-1}$. Except for the control, the other treatments had leaf area index values above 4. In conclusion, EC 2.0 $dS \cdot m^{-1}$ in grafted cucumber seedlings can improve growth, seedling quality, and chlorophyll concentrations.

Additional key words: chlorophyll concentration, compactness, electrical conductivity, nutrient solution, plug seedling production

Introduction

Cucumber is an important vegetable with global production reaching 94.72 million tons in 2022 (FAO, 2024). Notably, 88.4% of the cultivated areas are located in Asia, with global production continuously increasing every year (FAO, 2024). Cucumber has unisexual flowers, and its flower buds differentiate from the inside of the axillary bud, not the growing point (Lee et al., 2021). Sexual differentiation in cucumber flowers is highly dependent on genetic traits and cultivars, as well as external factors such as temperature, light, and amount and quality of pollination (Alotaibi et al., 2024). The cucumber is an annual herbaceous plant and is a cool season crop among fruits and vegetables (Smitha and Sunil, 2017). Therefore, cucumber is cultivated by using grafted seedlings to improve growth ability at low temperatures, prevent soil infectious diseases, and extend the harvest period (Lee and Oda, 2003). Grafted cucumber seedlings account for 75% of cucumber production in Korea (Lee et al., 2010a), highlighting the very high utilization rate of grafted cucumber plug seedlings (Moon et al., 2010).

Plug seedlings are known to save time and labor and provide the right environment for each stage of growth, resulting in healthy seedlings (Jeong, 1998). In the process of producing these plug seedlings, operations such as

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growing medium preparation and filling, sowing, fertilization, and environmental management are consistently systematized for standardization, efficiency, stabilization, and year-round planned production (Kim, 2015). The year-round production of standardized seedlings in a device-sized seedling production facility is referred to as plug-seedling production (Kim, 2015). Several studies have researched the effects of management and various treatments on the growth of seedlings using plug trays. For example, Jeong et al. (2020a) investigated the appropriate EC levels and management methods during the commercial nursery of *Astragalus membranaceus*. Shim et al. (2018) studied the composition and concentration of fertilizer suitable for the growth of pepper plug seedlings. Meanwhile, Moon et al. (2010) focused on saline stress treatment for suppressing stem elongation.

When mass-producing these plug seedlings, fertigation is used, which is a method of growing crops while simultaneously fertilizing and irrigating (Lee et al., 2010b). If the concentration of fertilizer in the medium falls below the appropriate level due to leaching or absorption by the plant, additional fertilization is required, and initial crop growth varies depending on when additional fertilization begins (Bunt, 1988). One of the things to consider when doing fertigation cultivation is the composition and concentration of the nutrient solution (Lee et al., 2010b). Nutrient solution is composed of what are called essential elements necessary for normal growth (Kang et al., 2022). Prior research, such as Kwack et al. (2015) and Yoo et al. (2012) have identified optimal nutrient solutions for various plants, thus showing the crucial influence of the type and composition of nutrient solutions on plant growth. In the nutrient solution, various inorganic elements exist in the form of ions, and their concentration can be measured through electrical conductivity (EC), indicating the ability of the solution to conduct electricity (Kang et al., 2022). Extremely low EC level of nutrient solutions generally leads to growth inhibition (Savvas and Adamidis, 1999). On the other hand, extremely high EC of nutrient solution causes osmotic stress, ionic toxicity, and growth restriction (Savvas and Adamidis, 1999). This mechanism has different effects on different crops, with tomatoes showing increased sugar and lycopene content at higher concentrations of nutrient solution (Wu and Kuborta, 2008). Meanwhile, strawberries are known to promote flower bud initiation at lower concentrations of nutrient solution (Gallace et al., 2017; Lieten, 2002; Sarooshi and Cresswell, 1994). In cucumber, it has been reported that when high EC is applied during the seedling stage, the plant height tends to decrease (Chung et al., 2001), while leaf growth is inhibited (Chung and Choi, 2001).

This study aims to investigate the impact of the EC levels of nutrient solution on the growth and seedling quality of grafted cucumber seedlings.

Materials and Methods

1. Plant material and growth conditions

Cultivars of cucumber (*Cucumis sativus* L.) 'NakWon-Seongcheongjang (Wonnongseed Co. Anseong, Korea), NWS', 'Sinsedae (FarmHannong Co. Ltd., Seoul, Korea), SD', and 'Goodmorning backdadagi (Nongwoo Bio Co. Ltd., Suwon, Korea), GB' were used as scions for grafting. Figleaf gourd (*Cucurbita ficifolia* 'Heukjong', FarmHannong Co. Ltd., Seoul, Korea) was used as the rootstock. The rootstock was sown on September 3, 2023, and the scion seeds were sown on October 5, 2023, both using the growing media (Pro-100, Cham Grow Co. Inc., Hongseong, Korea). The scions and rootstock were grafted together on October 18, 2023, using the root pruning-single cotyledon splice grafting method. After graft-taking, cucumber seedlings in plug trays were moved to a closed-type plant production system (CPPS, C1200H3, FC Poibe Co. Ltd., Seoul, Korea).

During the nursery period, the growth conditions within the CPPS were monitored using a temperature and humidity logger (TR-76Ui, T&D Co., Ltd. Matsumoto, Japan), with the change in temperature and relative humidity measured as shown in Fig. 1. The day temperature was set to 25°C and the night temperature to 15°C. The measured average day and night temperature were 26 ± 0.1°C and 15 ± 0.1°C, and average day and night relative humidity was 60 ± 0.5% and 70 ± 0.5%. The light quality was set as specified in Fig. 2. The light intensity was measured at the growing point using a spectrometer (HD2101.2, Delta Ohm S.r.L., Caselle, Italy), and it was confirmed to be maintained at 300 ± 20 $\mu mol \cdot m^{-2} \cdot s^{-1}$.

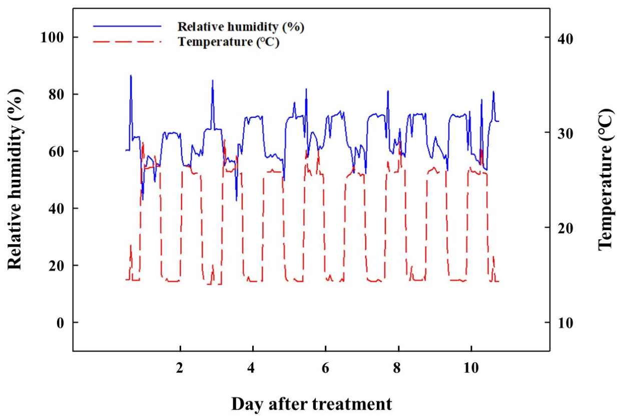

Fig. 1. The changes in temperature and relative humidity during 12 days after treatment in a closed-type plant production system.

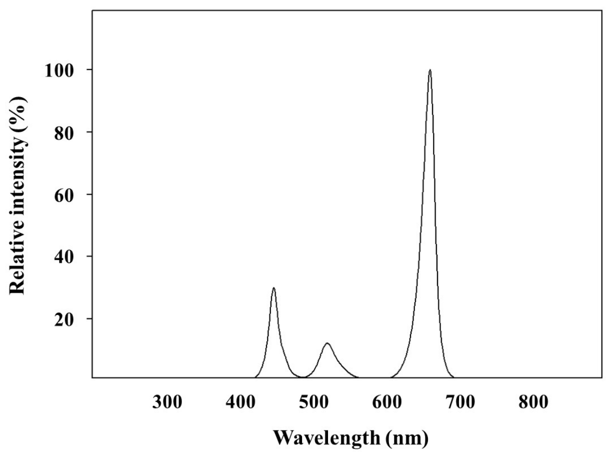

Fig. 2. The relative spectral distribution of light-emitting diodes in a closed-type plant production system.

2. EC level treatment of nutrient solutions

The nutrient solution used in the experiment utilized a commercial NPK fertilizer (YaraTera Kristalon Yellow, Yara International ASA, Oslo, Norway). The EC levels were set to 0.2 (control, 0.2 $dS \cdot m^{-1}$), 1.5 (EC 1.5 $dS \cdot m^{-1}$), 2.0 (EC 2.0 $dS \cdot m^{-1}$), and 2.5 (EC 2.5 $dS \cdot m^{-1}$). The EC of tap water was controlled at 0.2 $ds \cdot m^{-1}$ and set as the control. The nutrient solution was supplied once daily with sub-irrigation at onset of light irradiation. The EC levels were controlled using a pH/EC meter (HI-98130, Hanna Instruments Inc., Woonsocket, RI, USA). When graft-taking was completed, the level of EC treatment started on October 25, 2023, and ended on November 3, 2023, for a total of 10 days.

3. Measurement of chlorophyll concentration and chlorophyll fluorescence

In this experiment, 80% acetone was used as the solvent for chlorophyll extraction, and chlorophyll was extracted from the grafted seedlings at 10 days after treatment. Samples were collected from the same location on the leaf using a chlorophyll punch. The samples were immersed in 10 mL of 80% acetone, and kept in the dark for 24 hours. The absorbance was subsequently measured at 663 nm and 645 nm using a spectrophotometer (Libra-S22, Biochorm Ltd., London, UK) (Kozukue and Friedman, 2003; Lichtenthaler, 1987). Next, the measured absorbance was substituted into Eqs. 1 and 2 to calculate the concentration of chlorophyll a and b.

$$\text{Chlorophyll a} = 12.72 \times A_{663} - 2.58 \times A_{645} \tag{1}$$

$$\text{Chlorophyll b} = 22.88 \times A_{645} - 5.50 \times A_{645} \tag{2}$$

For the analysis of chlorophyll fluorescence (Fv/Fm), a portable chlorophyll fluorescence meter (PAM-2100, Heinz Walz GmbH, Effeltrich, Germany) was used to measure the chlorophyll fluorescence response characteristics of cucumber grafted seedlings. Fv/Fm was measured after the EC level treatments, in three replicates of two seedlings per plug tray, for a total of six seedlings.

4. Growth characteristics and seedling quality

The seedlings were randomly selected from uniform individuals in each tray. Plant height, hypocotyl length, stem diameter, soil plant analysis development (SPAD) value, leaf length, leaf width, number of leaves, leaf area, and fresh and dry weights of shoot and root were measured. Plant height and hypocotyl length were measured from the soil surface up to a growing point, and to a grafting point, respectively. Stem diameters were measured using a digital vernier calipers (CD-20PX, Mitutoyo Co. Ltd., Kawasaki, Japan), and chlorophyll concentrations were expressed as the SPAD value obtained using a portable chlorophyll meter (SPAD-502, Konica Minolta Inc., Tokyo, Japan). The number of leaves was measured as leaves larger than 1 cm, and leaf area was measured using a leaf area meter

(LI-3000, LI-COR Inc., Lincoln, NE, USA). Fresh and dry weights of shoot and root were measured using an electronic scale (EW220-3NM, Kern&Sohn GmbH., Balingen, Germany), and dry weight samples were dried in a constant temperature-drying oven (Venticell-222, MMM Medcenter Einrichtungen GmbH., Planegg, Germany) at 70°C for 72 hours before measurement. For the seedling quality analysis, compactness, leaf area rate (LAR), specific leaf area (SLA), leaf area index (LAI), and dry matter were calculated according to the following equation (RDA, 2012).

(a) Compactness ($mg \cdot cm^{-1}$) = (shoot dry weight) / (plant height)
(b) Leaf area rate ($cm^2 \cdot g^{-1}$) = (leaf area) / (total dry weight)
(c) Specific leaf area ($cm^2 \cdot g^{-1}$) = (leaf area) / (leaf dry weight)
(d) Leaf area index = (leaf area) / (cultivated area)

5. Statistical analysis

Treatments were laid out in a randomized complete block design in triplicate. Each treatment consisted of 18 plants. Six plants were used for each replicate. Statistical analyses were performed using the Statistical Analysis System (SAS 9.4; SAS Institute Inc., Cary, NC, USA). The experimental results were subjected to analysis of variance (ANOVA) and Duncan's multiple range tests. Significant differences were considered at $p \leq 0.05$. Graphs were plotted using the SigmaPlot software package (SigmaPlot 14.5, Systat Software Inc., San Jose, CA, USA).

Results and Discussion

1. Growth characteristics

Growth of grafted cucumber seedlings treated with EC levels was measured at 10 days after treatment (Fig. 3). There was no significant difference in the stem diameter of grafted cucumber seedlings (Table 1). In the EC 1.5 $dS \cdot m^{-1}$, both plant height and the hypocotyl length of the GB were significantly increased compared to the control. However, shoot fresh and dry weights tended to significantly increase

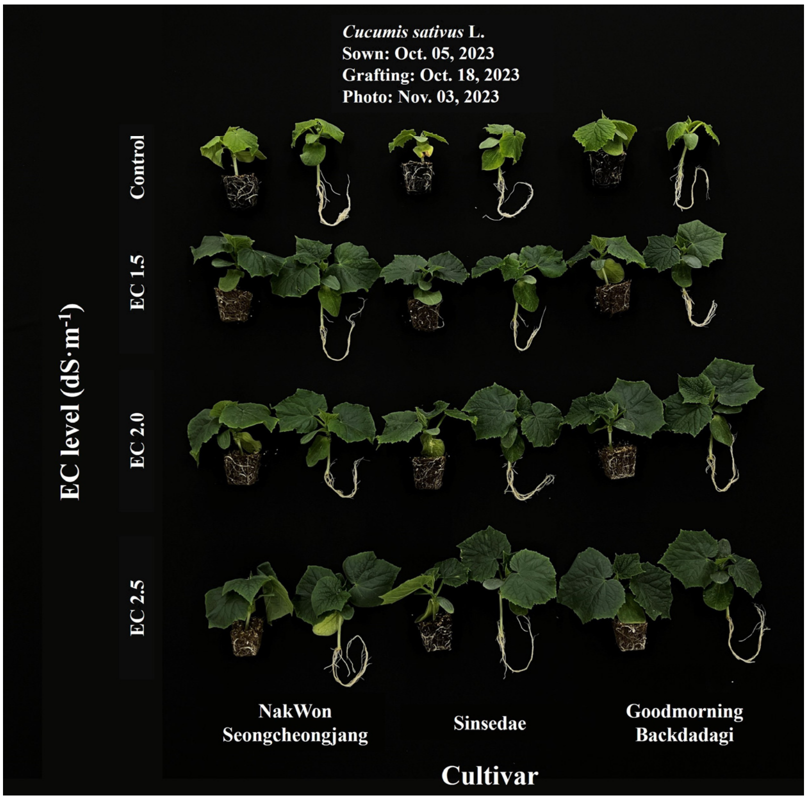

Fig. 3. Growth of grafted cucumber seedlings as affected by the different EC level and cultivar at 10 days after treatment.

Table 1. Shoot growth characteristics of cucumber seedlings as affected by the different EC level and cultivar at 10 days after treatment ($n = 6$).

EC level (dS·m^{-1}) (A)	Cultivar[z] (B)	Plant height (cm)	Hypocotyl length (cm)	Stem diameter (mm)	Fresh weight (g·plant^{-1})	Dry weight (g·plant^{-1})
Control	NWS	7.33 c[y]	5.35 a-c	3.93	3.29 d	0.33 bc
	SD	7.23 c	4.87 c	3.96	3.32 d	0.32 c
	GB	7.25 c	4.98 bc	4.04	3.38 d	0.33 bc
1.5	NWS	8.25 a	5.60 ab	4.12	5.00 b	0.38 a
	SD	8.03 ab	5.20 a-c	3.93	4.46 c	0.37 ab
	GB	7.75 bc	5.62 a	4.05	4.81 bc	0.41 a
2.0	NWS	7.55 bc	5.18 a-c	3.94	5.75 a	0.41 a
	SD	7.67 bc	5.63 a	4.22	5.26 b	0.40 a
	GB	7.42 c	5.58 ab	4.16	5.80 a	0.41 a
2.5	NWS	7.67 bc	5.57 ab	3.91	6.15 a	0.42 a
	SD	7.42 c	4.93 c	3.93	5.23 b	0.39 a
	GB	7.42 c	5.57 ab	3.93	5.77 a	0.40 a
Significant	A	***	*	NS	***	***
	B	NS	NS	NS	***	NS
	A×B	NS	NS	NS	NS	NS

[z]NWS, NakWonSeongcheongjang; SD, Sinsedae; and GB, Goodmorning backdadagi.
[y]Mean separation within columns by Duncan's multiple range test at $p \leq 0.05$.
NS, *, *** Nonsignificant or significant at $p \leq 0.05$, or 0.001, respectively.

at the EC 2.0 and 2.5 dS·m^{-1} compared to the control in the GB and NWS. When crops are exposed to salinity, changes occur in their morphological, physiological, and metabolic processes, so growth is inhibited and transpiration is suppressed by reducing stem elongation (Navetiyal et al., 1989). However, an exception was noted in a study by Huang et al. (2010), which reported that using figleaf gourd as a rootstock can alleviate growth inhibition, allowing plants to thrive even at high EC concentrations. This study determined that using figleaf gourd as a rootstock enhances resistance to growth inhibition caused by high EC, resulting in normal growth.

2. Leaf characteristics

After 10 days of treatment, leaf length and width tended to increase in all cultivars compared to the control, although there were differences among cultivars (Table 2). In the NWS, leaf length and width increased significantly with the increasing EC levels. SD showed a tendency to increase at the EC 2.0 and 2.5 dS·m^{-1} in all parameters. Leaf length and width of GB were significantly increased at the EC 2.0 and 2.5 dS·m^{-1} compared to the control. Significant increases in leaf length and leaf width affected the value of leaf area. Leaf area was significantly increased at the EC 2.0 and 2.5 dS·m^{-1} in all cultivars. The number of leaves was significantly increased in all treatments of all cultivars except the control. Jeong et al. (2016) reported that cucumber seedlings are optimal for shipping when they have 4－5 leaves, and in this experiment, all treatments were able to advance shipping more than the control. Furthermore, leaf fresh weight was increased significantly with the increasing EC, suggesting that an increase in leaf area and leaf fresh weight contributed to the increase in fresh weight of shoot.

3. Chlorophyll concentrations and fluorescence

SPAD was significantly increased in the treatments of all cultivars compared to the control (Fig. 4A). Chlorophyll b increased in the EC 2.0 and 2.5 dS·m^{-1}, and chlorophyll a increased in the treatment of all cultivars compared to the control (Figs. 4B and 4C). Therefore, chlorophyll concentration per unit of leaf area was significantly increased in the EC 2.0 and 2.5 dS·m^{-1}. Nitrogen is often the most

Table 2. Leaf characteristics of cucumber seedlings as affected by the different EC level and cultivar at 10 days after treatment ($n = 6$).

EC level ($dS \cdot m^{-1}$) (A)	Cultivar[z] (B)	Number of leaves	Leaf length (cm)	Leaf width (cm)	Leaf area ($cm^2 \cdot plant^{-1}$)	Leaf fresh weight (g)
Control	NWS	3.0 b	4.52 d[y]	6.78 e	41.9 e	1.1 e
	SD	3.0 b	4.35 d	6.00 f	37.7 e	1.2 e
	GB	3.0 b	4.92 d	6.62 ef	40.4 e	1.3 e
1.5	NWS	4.0 a	6.92 bc	8.65 bc	99.5 b	2.6 b
	SD	4.0 a	6.32 c	7.73 d	82.4 d	2.1 d
	GB	4.0 a	6.90 bc	8.62 bc	95.1 bc	2.2 cd
2.0	NWS	4.0 a	7.38 ab	9.27 ab	111.2 a	2.7 b
	SD	4.0 a	6.67 c	8.52 bc	92.2 c	2.5 bc
	GB	4.0 a	7.53 ab	9.95 a	108.1 a	2.7 b
2.5	NWS	4.0 a	7.73 a	9.53 a	113.4 a	3.2 a
	SD	4.0 a	6.55 c	8.42 cd	90.9 c	2.6 b
	GB	4.0 a	7.53 ab	9.53 a	111.2 a	2.8 ab
Significant	A	***	***	***	***	***
	B	***	***	***	***	**
	A×B	***	NS	NS	**	NS

[z]NWS, NakWonSeongcheongjang; SD, Sinsedae; and GB, Goodmorning backdadagi.
[y]Mean separation within columns by Duncan's multiple range test at $p \leq 0.05$.
NS, **, *** Nonsignificant or significant at $p \leq 0.01$, or 0.001, respectively.

limiting mineral element in plants and the crucial component of chlorophyll (Sinha, 2000). Increasing the EC level of a nutrient solution increased the concentration of inorganic nutrients, so the nitrogen content also increased. As a result, it was determined that chlorophyll concentration increases at the EC 2.0 and 2.5 $dS \cdot m^{-1}$, where inorganic nutrient content is higher.

The Fv/Fm measured in grafted cucumber seedlings were between 0.78-0.85, which represents no stress conditions (Gorbe and Calatayud, 2012) (Fig. 5). Fv/Fm indicates the stress levels caused by abiotic and biotic stressors (Baker and Rosenqvist, 2004). Fv/Fm is measured by irradiating dark-acclimated leaves with light and measuring the minimum fluorescence 'Fo', followed by irradiation with light at saturation levels to measure the dark-acclimated maximum fluorescence 'Fm'. The difference between maximum and minimum fluorescence is called variable fluorescence 'Fv', and the relative magnitude of vadiable fluorescence to maximum fluorescence, Fv/Fm, is the maximum quantum yield, the value of which was measured (Bussotti et al., 2020; Shin et al., 2020). In the case of abiotic factors, such as salt stress, an appropriate level can enhance plant productivity (Rouphael et al., 2018), but excessive salt concentration can reduce plant growth and yield (Adhikari et al., 2019). Thus, it is suggested that the cucumber seedlings did not undergo stress under any of the treatments.

4. Seedling quality

Fig. 6 shows the seedling quality of grafted cucumber seedlings as affected by the EC levels of the nutrient solution. Regarding compactness, there was a significant increase in the EC 2.0 and 2.5 $dS \cdot m^{-1}$ for all cultivars (Fig. 6A). Similarly, the LAR showed that all treatments increased compared to the control, especially the EC 2.0 and 2.5 $dS \cdot m^{-1}$ (Fig. 6B). SLA increased as the EC level increased, and showed a significantly higher tendency especially at the EC 2.5 $dS \cdot m^{-1}$ (Fig. 6C). Seedlings with superior quality can improve production stability and efficiency, being generally known as vigorous seedlings with high assimilation ability (RDA, 2008). Seedling compactness is the ratio of shoot dry weight and the plant height, and high compactness means the seedling is shorter

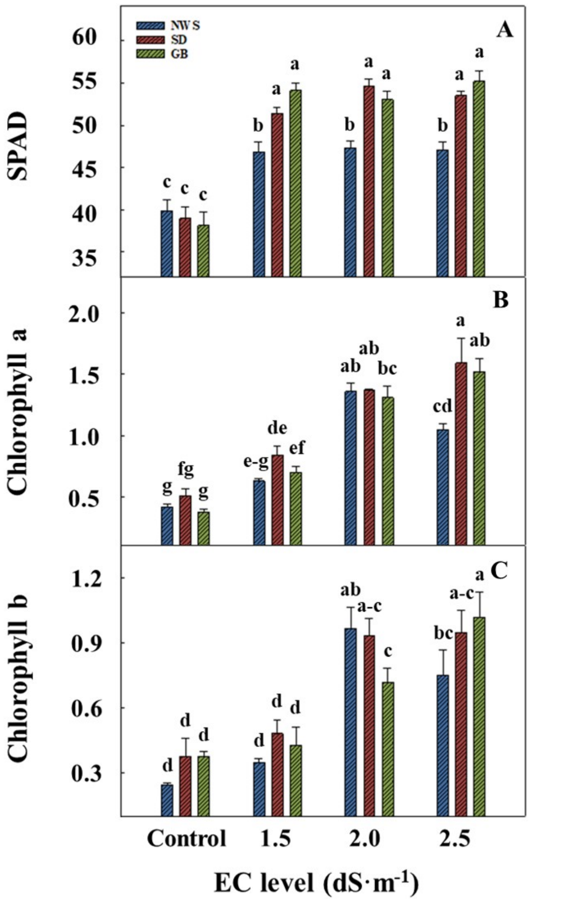

Fig. 4. SPAD (A), chlorophyll a (B), and chlorophyll b (C) of grafted cucumber seedlings as affected by the different EC level and cultivar at 10 days after treatment. Vertical bars indicate standard errors of the means (n = 6). Duncan's multiple range test at $p \leq 0.05$.

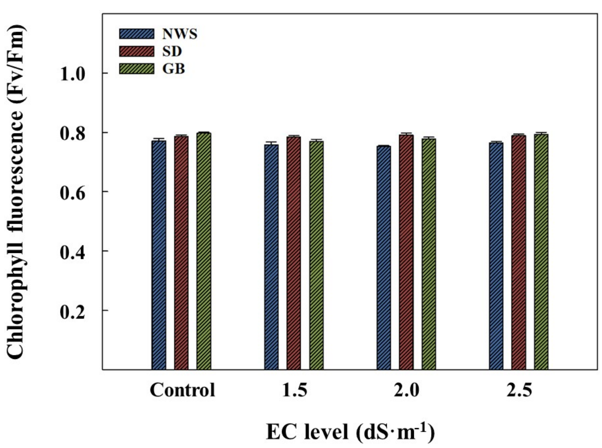

Fig. 5. Chlorophyll fluorescence (Fv/Fm) of grafted cucumber seedlings as affected by the different EC level and cultivar at 10 days after treatment. Vertical bars indicate standard errors of the means (n = 6). Duncan's multiple range test at $p \leq 0.05$.

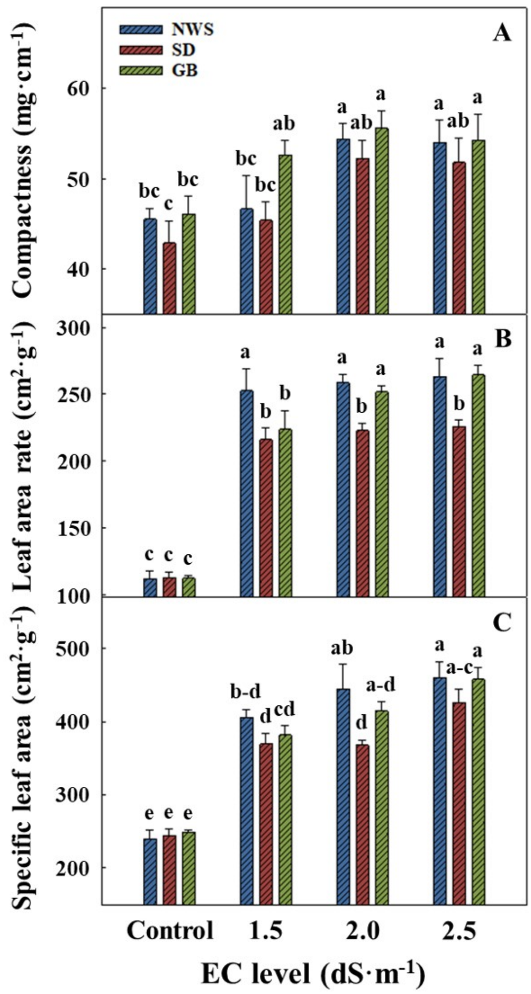

Fig. 6. Compactness (A), leaf area rate (B), and specific leaf area (C) of grafted cucumber seedlings as affected by the different EC level and cultivar at 10 days after treatment. Vertical bars indicate standard errors of the means (n = 6). Duncan's multiple range test at $p \leq 0.05$.

and thicker (Jeong et al., 2020b). LAR is the ratio of leaf area to the total fresh weights, and a measure of photosynthetic machinery per unit of plant biomass (Amanullah et al., 2007). SLA indicates of leaf thickness, with high values producing large, thin leaves and low values producing small, thick leaves (Louwerse and Zweerde, 1977). Previous studies have reported that when plants are stressed with high EC concentrations, both chlorophyll concentration and SLA can decrease (Choi and Lee, 2001; Lee et al., 1997). This phenomenon can be explained by the fact that as the EC levels increase, epidermal, descending, spongy, and lower epidermal cells shorten, resulting in

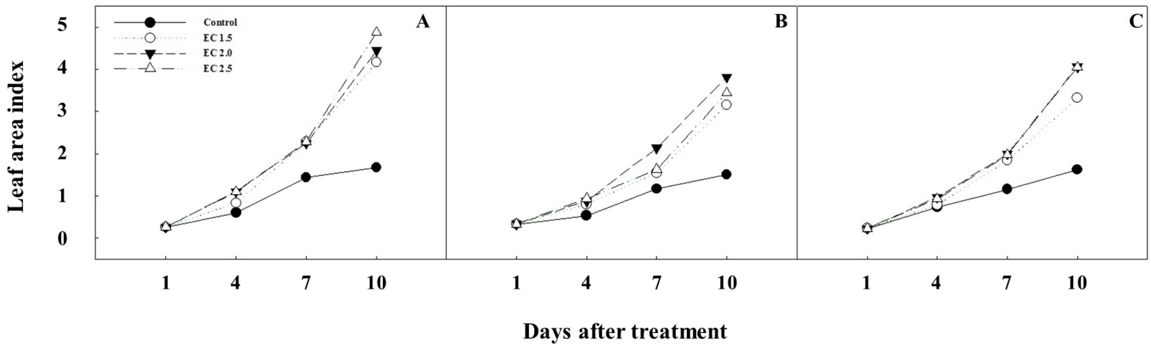

Fig. 7. Changes in leaf area index (LAI) of grafted cucumber seedlings over treatment days. Cultivars are NakWonSeongcheongjang (A), Sinsedae (B), and Goodmorning backdadagi (C).

lower SLA values (Chung and Choi, 2001).

LAI was significantly higher in all treatments compared to the control 10 days after treatment (Fig. 7). LAI is the primary growth metric associated with the growth and yield of a crop. Until the leaf area index reaches a certain value, it is important to have an optimal leaf area because it is correlated with photosynthesis and transpiration. However, as the leaf area index increases, the growth rate decreases due to shading between plant individuals or leaf-to-leaf (Jang et al., 2019). Goudriaan and Laar (1994) reported that exponential growth occurs until the leaf area index reaches 3, but above 3 the growth rate decreases. However, it has been reported that crops with a morphology like cucumber, which have relatively small upper leaves and an upright growth form, continue to grow normally even when the leaf area index is between 5 and 10 (Byun et al., 2014; Yoshida, 1981). As a result, it is considered that the plants can be more advanced shipping than the control when considered together with the number of leaves.

Conclusion

Shoot growth showed a significant increase at EC 2.0 and 2.5 $dS \cdot m^{-1}$ compared to other treatments in all cultivars. Compactness also increased significantly between EC 2.0 and 2.5 $dS \cdot m^{-1}$. When considering both the LAI and the number of leaves, all treatments could be advanced shipping. Since there was no significant difference in growth between the EC 2.0 and 2.5 $dS \cdot m^{-1}$, it was determined that the treatment with the lower EC is more suitable from the perspective of fertilizer consumption and horticultural practices. Therefore, for fertigation of grafted cucumber seedlings, fertilization with EC 2.0 $dS \cdot m^{-1}$ could produced high quality seedlings in all cultivars.

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EC 수준에 따른 품종별 오이 접목묘의 생육

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적 요. 식물 생육을 위한 양액은 물과 필수 무기 영양소를 포함하고 있다. 이러한 무기 영양소의 농도는 전기 전도도를 통해 측정할 수 있다. 전기 전도도(EC) 수준에 따라 서로 다른 작물이 각각 다른 생장 반응을 보이기 때문에 적절한 영양 용액 농도를 찾는 것이 중요하다. 따라서 본 실험은 다양한 품종을 고려하여 오이 접목묘 생산을 위한 적절한 EC 수준을 구명하기 위해 실시되었다. 접목에 사용된 오이(*Cucumis sativus* L.) 품종은 ‘낙원청장(NWS)’, ‘신세대(SD)’, ‘굿모닝 백다다기(GB)’의 세 가지이다. 대목은 흑종 호박(*Cucurbita ficifolia*)을 사용하였다. 이들은 2023년 9월 18일에 접목되었다. 접목 활착 후, 오이 접목묘는 플러그 트레이에서 밀폐형 식물 생산 시스템으로 옮겨졌다. 플러그 트레이의 오이 접목묘는 EC 1.5, 2.0, 2.5 $dS \cdot m^{-1}$로 처리되었다. 수돗물의 EC가 대조구로 사용되었다. 모든 품종에서 EC 2.0 및 2.5 $dS \cdot m^{-1}$에서 지상부의 생체중과 건물중이 유의미하게 증가하였다. 특히, NWS와 GB 품종의 엽장, 엽폭, 엽면적이 EC 2.0 및 2.5 $dS \cdot m^{-1}$에서 유의미하게 증가하였다. 묘소질 중 충실도와 엽면적 비율도 EC 2.0 및 2.5 $dS \cdot m^{-1}$에서 증가하였다. EC 수준이 증가함에 따라 비엽면적은 증가하였다. SPAD 값과 엽록소 a 및 b는 EC 2.0 및 2.5에서 유의미하게 높았다. 대조구를 제외한 다른 처리에서는 엽면적 지수 값이 4 이상이었다. 결론적으로, 접목된 오이 접목묘에서 EC 2.0 $dS \cdot m^{-1}$는 생장, 묘목 품질 및 엽록소 농도를 향상시킬 수 있다.

추가 주제어 : 엽록소 형광, 충실도, 전기 전도도, 양액, 공정육묘